CS264 Assignment 2
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1 Submission Instructions

This assignment is **18%**, please upload your **source code** to moodle before 23:59, 22nd-October. Please note that your source code has to be **C++**, otherwise your submission will be automatically rejected.

2 Plagiarism

Zero-tolerance policy with plagiarism is applied to this assignment. If your submission is found to be copied from someone else's work or machine generated or downloaded from internet. 0 mark will be given immediately and a case will be reported to the Department and University.

3 Late Submission

Late submissions will receive cumulative penalties to the awarded mark; 5% for first day, 15% for second day, 35% for third day, 65% for fourth day. Late submissions after the fourth day will not be accepted.

4 Writing your own String functions

In this assignment, you need to write your own string functions based on the following requirements **without using built-in C++ string**.

4.1 length

```
unsigned int length (const char* str)
```

[Description]: The `length` function shall return the length of a string excluding the `'\0'` character.

[Example]: `length("string")` returns 6 and `length("string\0")` also returns 6.

4.2 copy

```
char* copy (char* dest, const char* src)
```

[Description]: The `copy` function shall copy one string from `src` (including the `'\0'` character) to `dest` and return the copied string. The copy should happen if the size of the destination string is large enough to store the copied string, otherwise the function returns `NULL`.

[Example]: `copy(dest, "string")`, when printing `dest` it should print `string` on the screen, if `dest` has enough space to hold `"string"`.

4.3 indexOf

```
int indexOf (char c, const char* str)
```

[Description]: The `indexOf` function shall return the position of the first occurrence of `c` in a string `str`. If no such character `c` is found, then it shall return -1.

[Example]: `indexOf('i', "string")`, it should return 3. The call `indexOf('a', "string")` should return -1.

4.4 substring

```
char* substring (int i, int j, const char* str)
```

[Description]: The `substring` function shall return a substring of a given string `str`. The substring begins with the character at index `i` upto index `j` $-$ 1. This function shall make sure `i` and `j` are in the valid range, otherwise it shall return

`NULL`.
[Example]: `substring(0, 3, "string")` should return `"str"`.

4.5 replace

```
char* replace (char c, char p, const char* str)
```

[Description]: The `replace` function shall return a new string that replaces every occurrence of `c` with `p` in a given string `str`. If no such character `c` is found, then a copy of the string `str` is returned.

[Example]: `replace('i','x', "string")`, it should return `"strxng"`.

5 Comments

For each function above, you should write a block of comment to explain what a function does including its return values and arguments. The comments will be **10%** of this assignment.

6 Test Cases

For each function above, you should also consider to write some test cases in other functions with the following name style:

`[function]_name_test_case_[k]`

where `[function]` is a function name and `[k]` is the number of test case. For example, the 1st test case for function `substring` will be `substring_test_case_1`; The total number of test cases is depending on you. However, the rule here is you should have sufficient number of test cases to test each function. The test cases will be **20%** of this assignment.